THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Mathematics & Physics

Semester Two Examinations, 2023

PHYS4055 Atomic Physics and Quantum Optics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is an Open Book examination
•Any calculator permitted - unrestricted
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

* Open-book: Any additional written or printed material is permitted; material may also be annotated.

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x 14-Page Answer Booklet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are FOUR questions. Answer any THREE.

For Examiner Use Only

Question	Mark

Total _____

Question 1.

(10 marks)

Consider a single mode of the electromagnetic field in a nonlinear medium represented by the Hamiltonian:

$$\hat{H} = \hbar\left(\omega a^{\dagger}a + \lambda\left(a^{\dagger}a\right)^{p}\right),$$

where ω and λ are real parameters, and p is an integer.

a) *(2 marks)* What are the eigenstates and eigenvalues of H?

b) *(2 marks)* What is the time evolution, in the interaction picture, of an initial coherent state $|\alpha\rangle$ under the action of this Hamiltonian? Write your final expression in the number-state basis.

c) *(2 marks)* Show that, for even p, the coherent state in the interaction picture at time $t = \frac{\pi}{2\lambda}$ can be written as;

$$|\psi(t)\rangle = \tfrac{1}{2}[(1-i)|\alpha\rangle + (1+i)|-\alpha\rangle].$$

Hint: For integer n and even p;

$$\exp(-\frac{i\pi}{2}n^{p}) = \frac{1}{2}[1 + (-1)^{n} - i + i(-1)^{n}]$$

d) *(2 marks)* Using the state shown in part c), determine the correlation function $g^{(k)}$ for positive integers k.

e) *(2 marks)* Given your answer to d), to what extent can the state shown in c) be described as classical? Describe any features that are non-classical.

Question 2.

(10 marks)

Consider the electronic structure of the hydrogen atom. The normalised wavefunction of an electron in the state 1s (assuming a point nucleus) is given by;

$$\psi_{1s}(r) = \sqrt{\frac{1}{\pi a_0^3}}\, e^{-\frac{r}{a_0}}.$$

a) *(2 marks)* What radius from the nucleus corresponds to the maximum probability of finding the electron?

b) *(2 marks)* Calculate the expectation value $\langle r \rangle$ and $\langle \frac{1}{r} \rangle$ for the ψ_{1s} state. How do these results compare with the result found using the Bohr theory?

Hint: $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$.

Now consider the electronic structure of the helium atom.

c) *(2 marks)* Write down the Schrödinger equation for helium and explain the physical significance of each of the terms.

d) *(2 marks)* There are four possible spin states of the two electrons in the helium atom. Write the states down and identify whether they are symmetric or antisymmetric under exchange. Label the singlet and triplet states.

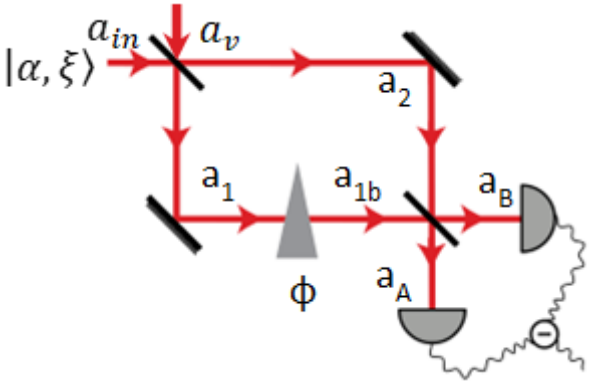
e) *(1 mark)* Explain the physical origin of hyperfine structure. Explain why the hyperfine structure is expected to be of orders of magnitude smaller than fine structure.

f) (1 mark) Consider the excited states of the Helium atom 2^1S_0, 2^1P_1, 2^3S_1 and $2^3P_{0,1,2}$. Draw the energy level diagram and list the possible electric-dipole allowed transitions.

Question 3.

(10 marks)

See the image below for an example interferometer.



a) (*2 mark*) The matrix $M = \begin{pmatrix} 1/\sqrt{2} & i/\sqrt{2} \\ i/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$ is used to represent the first beam splitter. Show that the output of the first beam splitter conserves energy when sending two input fields E_{in} and E_{v} into the input ports.

b) (*1 mark*) We now send a squeezed coherent state into this interferometer $|\psi_{\mathrm{in}}\rangle = |\alpha, \xi\rangle_{in}|0\rangle_v$ with squeezing parameter r and φ_S ($\xi = re^{2i\varphi_S}$). Do you expect a sensitivity enhancement due to the squeezing? Explain.

c) (*2 marks*) Knowing that $a_A = \frac{1}{2}\left[a_{in}\left(e^{i\phi}+1\right) + a_v\left(e^{i\phi}-1\right)\right]$ and $a_B = \frac{1}{2}\left[a_{in}\left(e^{i\phi}-1\right) + a_v\left(e^{i\phi}+1\right)\right]$ show that,

$$n_A - n_B = \cos(\phi)\,(n_{in} - n_v) + i\sin(\phi)\left(a_{in}^{\dagger}a_v - a_v^{\dagger}a_{in}\right).$$

d) (*2 marks*) Calculate the expression of the signal $\langle n_A - n_B\rangle$ for the squeezed coherent input state. Use the expressions $\mu = \cosh(r)$ and $\nu = e^{2i\varphi_S}\sinh(r)$, and their associated identities, to get the solution in terms of α and ν.

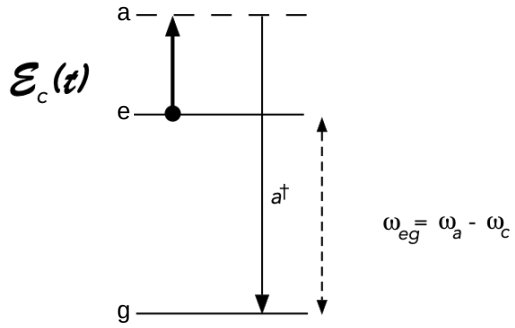
e) (*2 marks*) Calculate the expression of the noise $\mathrm{Var}[n_A - n_B]$ for a phase shift $\phi = \pi/2$. Use the expressions $\mu = \cosh(r)$ and $\nu = e^{2i\varphi_S}\sinh(r)$, and their associated identities, to get the solution in terms of α and ν.

f) (*1 marks*) Using the expression of the $\mathrm{SNR} = \langle n_A - n_B\rangle / \sqrt{\mathrm{Var}[n_A - n_B]}$, calculate the smallest resolvable change in phase shift with this squeezed state for ϕ around $\pi/2$.

Question 4.

(10 marks)

A two-photon Raman transition couples two atomic states $|g\rangle$ and $|e\rangle$ via an off-resonant transition with an auxiliary state, $|a\rangle$, as shown below.



The field $\mathcal{E}_c(t)$ is a time dependent classical control field at carrier frequency ω_c and $a, a^\dagger$ are the annihilation and creation operators of a quantised cavity field with resonant frequency ω_a. The diagram above shows the resonant case, but in general there can be a detuning $\delta = \omega_{eg} - (\omega_a - \omega_c)$ where ω_{eg} is the transition frequency between the ground and excited state. In the dipole and rotating wave approximation the Hamiltonian is

$$H = \hbar\omega_a a^\dagger a + \hbar\frac{\omega_{eg}}{2}\sigma_z + \hbar g\left(\mathcal{E}_c(t)a^\dagger|g\rangle\langle e|e^{-i\omega_c t} + \mathcal{E}_c^*(t)a|e\rangle\langle g|e^{i\omega_c t}\right).$$

We will ignore spontaneous emission in this question as the coupled states are dipole forbidden transitions.

a) *(2 marks)* Show that the interaction picture Hamiltonian is
$$H_I = -\hbar\delta a^\dagger a + \hbar g\mathcal{E}_c(t)\left(a^\dagger|g\rangle\langle e| + a|e\rangle\langle g|\right).$$

b) *(3 marks)* If the cavity field is strongly driven we may replace $a, a^\dagger$ by the semiclassical coherent amplitudes α, α^*. Show that the equations of motion for the atomic system are

$$\frac{dz(t)}{dt} = 2ig\mathcal{E}_c(t)(\alpha^* w^* - \alpha w)$$

$$\frac{dw(t)}{dt} = -ig\mathcal{E}_c(t)\alpha^* z$$

where $z = \langle\sigma_z\rangle, w = \langle\sigma_+\rangle, w^* = \langle\sigma_-\rangle$.

c) *(3 marks)* Define a pulse amplitude as
$$A = \int_{-\infty}^{\infty}\mathcal{E}_c(t)dt = \frac{2\mathcal{E}_0}{\gamma}$$
given the Lorentzian control pulse $\mathcal{E}_c(t) = \mathcal{E}_0 e^{-\gamma|t|}$.
Starting with the equations in b), find the conditions on $\mathcal{E}_0, \gamma, \delta$ such that an atom prepared in the ground state in the far past of the pulse will evolve to the excited state after the passage of the control pulse.

Hint: define a new time variable $\tau(t) = \int_{-\infty}^{t} \mathcal{E}_c(t')dt'$ such that $\frac{d\tau}{dt} = \mathcal{E}_c(t)$. Also, recognising the coherent amplitude α is dependent on detuning, use the simplified expression $\sqrt{2}g\sqrt{\alpha^2 + \alpha^{*2}} = \omega(\delta)$.

d) *(2 marks)* How might this process be used as a sensor for the detuning δ?

END OF EXAMINATION